

Uniformly bounded translations on de Branges spaces: a complete Paley–Wiener dichotomy

Full result for arXiv:1801.01345

13 August 2026

Status: full classification, likely valid. Every nonzero de Branges space whose full real-translation group is uniformly bounded is either a Paley–Wiener space as a set or the one-dimensional space of constants. Thus the intended infinite-dimensional converse is true; the constants are the unique exception if the source question is read literally without a dimension hypothesis.

1. Source question and answer

Baranov and Bommier-Hato prove that a de Branges space has a representing Fock weight depending only on the imaginary part exactly when it is a Paley–Wiener space PW_a , $a > 0$. They observe that this property implies uniform boundedness of all real translations

$$U_t F(z) = F(z + t), \quad t \in \mathbb{R},$$

and ask whether the converse holds [1].

The following theorem completely answers the question, including its finite-dimensional edge case.

Theorem 1 (Uniform-translation dichotomy). *Let $\mathcal{H}$ be a nonzero de Branges space and suppose that every U_t maps $\mathcal{H}$ onto itself and*

$$C := \sup_{t \in \mathbb{R}} \|U_t : \mathcal{H} \rightarrow \mathcal{H}\| < \infty.$$

Then exactly one of the following holds:

1. $\mathcal{H} = PW_a$ as a set, for some $a > 0$, and its norm is equivalent to the usual Paley–Wiener norm;
2. $\mathcal{H} = \mathbb{C}$ as a set.

In particular, every infinite-dimensional such space is a Paley–Wiener space. The second case is realized by $\mathcal{H}(z + i)$ and is the unique literal exception to the converse of Theorem 2.4 in the source.

2. Proof intuition

Uniform boundedness can first be removed by averaging the original inner product over longer and longer translation intervals. The resulting equivalent norm makes every translation unitary. The decisive point is that this averaging preserves all de Branges axioms: conjugation commutes with translations, while a zero-moving Blaschke factor at w is carried by a translation to the corresponding factor at $w - t$.

For the unitary norm, the reproducing kernel is invariant under simultaneous real translation of both variables. If $E = A - iB$ generates the space, the row $V = (A, B)$ must therefore satisfy

$$V(z + t) = V(z)M_t, \text{quad} M_t \in SL(2, \mathbb{R}).$$

2.3. Weights depending on imaginary part. We have seen that for the Paley–Wiener space PW_a , one can choose a representing Fock weight $W_0(z) = (|\operatorname{Im} z| + 1)^{-1} e^{-a|\operatorname{Im} z|}$ which depends only on $\operatorname{Im} z$. Are there other de Branges spaces with the same property? The answer turns out to be negative:

Theorem 2.4. *Let $\mathcal{H}(E)$ be a de Branges space. Then the following are equivalent:*

- (i) *There exists a representing weight W for $\mathcal{H}(E)$ such that $W(z) = W(\operatorname{Im} z)$, $z \in \mathbb{C}$;*
- (ii) *$\mathcal{H}(E) = PW_a$ for some $a > 0$.*

Note that there exist functions E such that $|E|$ is not equivalent to 1 on $\mathbb{R}$, but $\mathcal{H}(E) = PW_a$. We discuss such construction (from paper [17]) in Section 7.

We conclude with one question which we were not able to answer. Note that the property (i) implies, in particular, that all real translations $U_t : F \mapsto F(\cdot + t)$, $t \in \mathbb{R}$, are isomorphisms of $\mathcal{H}(E)$ onto itself and $\sup_{t \in \mathbb{R}} \|U_t : \mathcal{H}(E) \rightarrow \mathcal{H}(E)\| < \infty$. It would be interesting to know whether the converse is true, that is, whether there exist de Branges spaces where all real translations are isomorphisms with uniformly bounded norms which are different from the Paley–Wiener spaces.

2.4. Organization of the paper. In Section 3 we prove some auxiliary estimates for the distances to the level sets. In Section 4 we prove the main result of the paper, Theorem 2.2. In Section 5 we prove necessity of the condition $\varphi' \in L^\infty$ in Theorem 2.1 (its sufficiency follows almost immediately from Theorem 2.2). In Section 6 we give another construction of a representing weight for a general de Branges space which uses geometry of the spectral data for $\mathcal{H}(E)$. Finally, in Section 7 we prove Theorem 2.4 on the weights depending on the imaginary part only.

3. DISTANCES TO THE LEVEL SETS

In this section we prove inequality (2.6).

Figure 1: Theorem 2.4 and the ensuing open question on page 5 of arXiv:1801.01345.

The matrices form a continuous one-parameter group, hence $M_t = e^{tX}$ for $X \in \mathfrak{sl}(2, \mathbb{R})$. Cayley–Hamilton leaves elliptic, parabolic, and hyperbolic possibilities. They give respectively a sinc kernel, a constant kernel, and a hyperbolic-sinc kernel. The first is Paley–Wiener, the second is one-dimensional, and the third violates the two-point Cauchy–Schwarz inequality for positive kernels.

3. Averaging lemma

Write $JF = F^\#$ and, when $F(w) = 0$ for $w \notin \mathbb{R}$, put

$$R_w F(z) = \frac{z - \bar{w}}{z - w} F(z).$$

Lemma 1. *Under the hypotheses of Theorem 1, there is an equivalent Hilbert norm $\|\cdot\|_0$ on the same functions such that:*

- 1. *every U_t is unitary for $\|\cdot\|_0$;*
- 2. *J is an isometry for $\|\cdot\|_0$;*
- 3. *$\|R_w F\|_0 = \|F\|_0$ whenever $F(w) = 0$ and $w \notin \mathbb{R}$.*

Consequently the new norm is again a de Branges norm.

Proof. Since $U_t^{-1} = U_{-t}$,

$$C^{-1} \|F\| \leq \|U_t F\| \leq C \|F\|. \tag{1}$$

The scalar coefficient functions of U_t are continuous. Indeed this is clear when the second vector is a reproducing kernel, because $\langle U_t F, K_w \rangle = F(w + t)$; density of the kernel span and the uniform bound extend continuity to every second vector.

For $T > 0$, define the bounded positive operator A_T weakly by

$$\langle A_T F, G \rangle = \frac{1}{2T} \int_{-T}^T \langle U_t F, U_t G \rangle dt.$$

Equation (1) gives

$$C^{-2}I \leq A_T \leq C^2I.$$

Choose a weak-operator cluster point A as $T \rightarrow \infty$. Translating the averaging interval by a fixed s changes only two boundary intervals of total length $2|s|$; the integrand is uniformly bounded. Hence

$$U_s^* A U_s = A \quad (s \in \mathbb{R}).$$

Moreover $C^{-2}I \leq A \leq C^2I$. Thus

$$[F, G]_0 := \langle A F, G \rangle, \quad \|F\|_0^2 = [F, F]_0,$$

is an equivalent Hilbert norm and all U_s are unitary.

The original de Branges conjugation is isometric and $JU_t = U_t J$. Taking the same weak limit of averaged squared norms shows $\|JF\|_0 = \|F\|_0$. Likewise,

$$U_t R_w F = R_{w-t} U_t F.$$

When $F(w) = 0$, the function $U_t F$ vanishes at $w - t$, so the original de Branges axiom gives $\|R_{w-t} U_t F\| = \|U_t F\|$ for every t . Averaging and passing to the cluster point yields $\|R_w F\|_0 = \|F\|_0$. Point evaluations remain bounded because the norms are equivalent. These are precisely the de Branges axioms [2]. $\square$

4. The $SL(2, \mathbb{R})$ orbit

By Lemma 1, represent the newly normed space isometrically as $\mathcal{H}(E)$ for a Hermite–Biehler function $E = A - iB$, where $A^\# = A$ and $B^\# = B$. Set

$$V(z) = (A(z), B(z)).$$

The de Branges reproducing-kernel formula is

$$K(z, w) = -\frac{\det \begin{pmatrix} V(z) \\ V(\bar{w}) \end{pmatrix}}{\pi(z - \bar{w})}. \quad (2)$$

Because translations are unitary, $K(z + t, w + t) = K(z, w)$. The denominator in (2) is unchanged, so all the determinants in its numerator are unchanged. Choose real x_1, x_2 such that $V(x_1), V(x_2)$ are linearly independent. Such points exist, since otherwise A and B would be proportional and the kernel would vanish. There is a unique real matrix M_t with

$$V(x_j + t) = V(x_j) M_t \quad (j = 1, 2).$$

Preservation of their determinant gives $\det M_t = 1$. Comparing determinants against the two independent translated rows then gives

$$V(z + t) = V(z) M_t \quad (z \in \mathbb{C}). \quad (3)$$

The matrices depend continuously on t , and uniqueness in (3) gives $M_{s+t} = M_s M_t$. Therefore

$$M_t = e^{tX} \quad \text{for some } X \in \mathfrak{sl}(2, \mathbb{R}).$$

Putting $v = V(0)$ and using analytic continuation from real z in (3),

$$V(z) = v e^{zX}, \quad z \in \mathbb{C}. \quad (4)$$

The rows v and vX are independent; otherwise the orbit lies in one line and again $K \equiv 0$.

5. Classification and proof of the theorem

Since $\text{tr } X = 0$, Cayley–Hamilton says

$$X^2 = -(\det X)I.$$

Write $e^{uX} = c(u)I + s(u)X$. From (4), multiplication of both determinant rows in (2) by $e^{-\bar{w}X}$ gives

$$K(z, w) = \gamma \frac{s(z - \bar{w})}{z - \bar{w}}, \quad \gamma = K(0, 0) > 0, \quad (5)$$

where $s'(0) = 1$ and exactly one of the following occurs:

$$\begin{array}{l|l} \det X = a^2 > 0 & s(u) = \sin(au)/a \\ \det X = 0 & s(u) = u \\ \det X = -a^2 < 0 & s(u) = \sinh(au)/a. \end{array}$$

In the elliptic case, (5) is a positive scalar multiple of the standard Paley–Wiener kernel

$$\frac{\sin(a(z - \bar{w}))}{\pi(z - \bar{w})}.$$

Hence the newly normed space, and therefore the original space as a set, is PW_a .

In the parabolic case the kernel is the constant γ , so the RKHS is exactly the one-dimensional space of constants. It is realized as the de Branges space $\mathcal{H}(z + i)$.

The hyperbolic case is impossible. For distinct real x, y ,

$$K(x, x) = K(y, y) = \gamma, \quad |K(x, y)| = \gamma \frac{\sinh(a|x - y|)}{a|x - y|} > \gamma,$$

contradicting $|K(x, y)|^2 \leq K(x, x)K(y, y)$ for a positive kernel. This proves Theorem 1.

6. Verification, novelty, and scope

The proof was checked at four structural points: the two-sided bounds survive the weak-operator limit; averaging preserves the zero-moving axiom because $U_t R_w = R_{w-t} U_t$; determinant invariance produces a genuine continuous $SL(2, \mathbb{R})$ group; and the hyperbolic kernel has a negative two-point Gram determinant. The three matrix-exponential and kernel forms were also checked symbolically by `code/sl2.kernel.audit.py`. This code is a mechanical audit only; no computational assertion is needed for the proof.

The classification distinguishes the source’s two possible readings. In the intended infinite-dimensional setting, the missing implication is true and, by the source’s Theorem 2.4, a representing weight depending only on the imaginary part exists. Without a dimension condition, $\mathcal{H}(z + i) = \mathbb{C}$ is a counterexample to that implication, but the theorem proves it is the only one.

The run indexes contained no duplicate. Bounded searches through 13 August 2026 covered the exact question, its title and authors, translation-invariant and periodic de Branges spaces, Bellavita’s 2021 and 2023 translation-operator papers, the 2024 homogeneous-space classification, and the 2026 composition-operator paper arXiv:2604.06801. The later works give criteria for individual translations or other group actions, but no resolution of this uniformly bounded full-group question was found. Novelty outside this bounded search is not asserted pending expert review.

Human-review recommendation. Verify the weak-operator averaging step, the preservation of the de Branges zero axiom, and the elementary passage from determinant invariance to $M_t \in SL(2, \mathbb{R})$. Also preserve the explicit one-dimensional caveat when stating the answer to the source.

References

- [1] A. Baranov and H. Bommier-Hato, *De Branges spaces and Fock spaces*, Complex Variables and Elliptic Equations 63 (2018), 907–930; arXiv:1801.01345.
- [2] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice–Hall, Englewood Cliffs, 1968.